

IT362-Data science

Project Report

Semester-2 1447H

Online Course Popularity & Rating Analysis

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Phase 1

1. Introduction

Problem Summary

Over the last decade, the e-learning market has grown largely, with platforms offering a wide range of courses across many fields to millions of learners worldwide. In the Arab region, platforms like Shaguf have emerged to offer courses specifically to Arab learners, making education more accessible. As the competition between online learning platforms increases, it's become important to understand how pricing strategies and discounts affect course popularity and

student enrollment. When choosing online courses, students take many factors into account, including price, course duration, and available discounts. While it seems logical that longer courses should come with higher prices and that discounts would appeal to price sensitive learners, the way these elements interact and influence each other is still not well understood especially on Arabic platforms. There has been little research into how factors like course length, final price, and discount percentages impact the place of courses on regional platforms.

Research Question

To address this gap, the present study looks to answer the following research question:

To what extent do course duration, final price, and discount percentage influence student enrollment on the Shaguf platform?

Research Approach

In this study, course price is defined as the final price after any discounts, while the discount percentage is calculated by comparing the original price with the current price. Course duration will be extracted from the course descriptions and converted into numerical hours for analysis. The research uses structured data such as price, original price, discount percentage, duration, number of students, and course level collected from publicly available course pages on the Shaguf platform through web scraping.

By analyzing recent data from Shaguf, this study aims to provide actionable insights that present a balance between offering affordable learning options for students and maintaining consistent revenue for course providers. In the end, this research looks to contribute to the growth and accessibility of online education within the Arab world.

2. Literature Review

This literature review provides an overview of studies on pricing strategies in online education, particularly in MOOCs (Massive Open Online Courses). MOOCs are online courses that allow an unlimited number of participants, usually offering open access to education across the globe. The focus is on how pricing strategies influence learner behavior, enrollment patterns, and overall market outcomes. In this section, we analyze several key studies that explore different facets of pricing strategies in MOOCs and online education products.

Study 1: Dynamic Pricing in Online Education Chen Qi (2025)

This study investigates the role of price in MOOCs, specifically examining how price functions as a signaling mechanism that influences learners' decisions to enroll. Although pricing plays an important role in education markets, there has been limited research on how various pricing strategies affect learner behavior in the context of MOOCs. The authors used web-crawled data from the edX platform to study course pricing, institutional affiliations, subjects, course durations, and enrollment patterns. The study found that price was a significant factor in enrollment decisions, even outweighing other factors like the reputation of the institution

providing the course or the subject matter of the course. This research offers valuable insight into the role of pricing strategies in attracting learners and promoting global educational accessibility.

Study 2: Research on Pricing Strategy of Online Education Products

Zhang, Gu, & Qu (2022)

The second study focuses on the challenges that online education companies face when determining their pricing strategies. It addresses the issue of uncertainty around consumers' perceived value of education products. The authors propose a theoretical model based on consumer utility functions, which divides consumers into three categories: matching consumers, price sensitive consumers, and indifferent consumers. The study compares two pricing strategies: price undertaking and intertemporal pricing. They found that the matching degree of consumers (how well their needs align with the course offering) and price sensitivity very much influence both product pricing and enterprise income. The intertemporal pricing strategy was more effective when the price sensitivity coefficient was high. This study helps the existing literature on online education pricing.

Study 3: Studying Learner Behavior with Free-Certificate Coupons

Littenberg-Tobias, Ruipérez-Valiente, & Reich (2020)

This study examines the effect of price trimming through free certificate coupons on learner behavior in MOOCs. The researchers used case studies to compare enrollment and completion rates between courses offering free certificates and those that did not. The results showed that offering free certificates increased enrollment in the verified certificate tracks and helped completion rates among participants. However, the availability of free certificates also led to lower certification rates in some cases. This shows that while price motivation can encourage enrollment, they may not necessarily increase completion rates in the long term, depending on how the price motivation is structured and communicated.

Study 4: Dynamic Pricing in Online Education – A Game-Theoretic Approach

Chen (2025)

This study uses a game theoretical approach to analyze the relation pricing of online courses, focusing on the conflict between the pricing strategies of the platform and the instructor. The study highlights how a platform's income maximizing pricing strategy would reduce instructor earnings, even though flexible pricing could potentially benefit both parties. Using a differential game model, the study analyzes three different pricing scenarios basic, static, and fully dynamic and explores the effects of instructors' quality and learners' reference prices.

The findings show that while dynamic pricing can maximize short term income for the platform, it may reduce long-term instructor withhold. To address this, the study suggests that platforms need to balance pricing strategies with fair revenue sharing agreements that protect instructors.

2. Differences in Modeling Approaches

- 1. Study 1:** This study employs a systematic approach, analyzing web-scraped data to understand MOOC pricing structures. It focuses on how price impacts enrollment decisions and compares it to other course attributes like subject, institution, and duration.
- 2. Study 2:** This study uses a theoretical model based on consumer useable functions, considering different consumer types (matching, price sensitive, and indifferent). No empirical data is included, as it relies on constructed models to analyze how consumer behavior and pricing strategies affect demand and income.
- 3. Study 3:** This study uses case studies to explore how price trimming, particularly through free certificate coupons, influence learner behavior in real-world MOOCs. It compares participation, certification rates, and learner engagement across different pricing strategies.
- 4. Study 4:** Employing a game-theoretic approach, this study models the strategic interactions between platforms, instructors, and learners under dynamic pricing conditions. It explores different pricing scenarios and introduces market uncertainty to analyze outcomes.

3. Strengths and Limitations of Existing Work

1. Study 1

Strengths: Provides valuable, data-driven insights into how pricing influences enrollment, helping understand the practical aspects of pricing strategies in MOOCs.

Limitations: The research is based on a single platform (edX), which may limit its generalizability to other platforms or markets.

2. Study 2

Strengths: Theoretical framework provides insights into how consumer behavior and pricing strategies interact.

Limitations: The absence of empirical data reduces its real-world applicability, making it harder to validate the model's predictions.

3. Study 3

Strengths: Real-world case studies provide concrete evidence of how price incentives like free certificates impact learner behavior.

Limitations: The study's reliance on only two case studies limits the ability to generalize findings, and it doesn't account for external factors that might affect learner decisions.

4. Study 4

Strengths: The use of game theory allows for a deep analysis of pricing dynamics and the interactions between platform and instructor strategies.

Limitations: As no empirical data is used, the findings may not fully reflect real world behaviors, and assumptions in the model could limit its relevance.

4. What is Missing, Limited, or Unclear in Previous Work

1. Study 1

More research is needed to consider additional factors, such as course quality and instructor reputation, that could influence pricing strategies and learner enrollment decisions.

2. Study 2

Theoretical models would benefit from empirical validation to assess how well they reflect actual consumer behavior in different markets.

3. Study 3

Broader studies with more case studies and the including of external factors could help generalize the findings to a bigger range of MOOCs and platforms.

4. Study 4

Empirical validation is necessary to see how learners actually respond to dynamic pricing strategies in practice, especially when market conditions turn.

5.1. How Our Project Builds Upon Existing Studies:

- 1. Focus on Pricing and Learner Behavior** Our project builds on studies like Shi et al. (2024) and Zhang et al. (2022), which examined pricing strategies and their effects on learner decisions. By using real-world data from the Shaguf platform, our project aims to offer deeper insights into how discounts, course duration, and other pricing factors influence enrollment decisions, especially in the Arabic market. This regional focus better helps the global understanding of pricing in MOOCs.
- 2. Empirical Evidence from a Regional Platform** While previous studies often rely on global platforms like edX and Coursera or theoretical models, our project uses real world data from Shaguf, an Arabic e-learning platform. This focus allows us to explore pricing dynamics specific to the Arabic market, which has not been presented a lot in prior research.
- 3. Expanded Dataset and Diverse Variables** Unlike studies that use limited datasets or focus on one platform, our project includes data from nearly ~1,000 courses on Shaguf. This inclusive dataset covers various categories, levels, and subjects (technology, business, human), allowing for a more accurate analysis of pricing strategies, discounts, and consumer behavior across different course types.

5.2. Differences from Existing Studies:

1. **Cross-Sectional Snapshot** Our dataset is cross-sectional, capturing pricing and discount patterns at a specific point in time (February 2026). While this limits the ability to analyze long-term trends, it provides a valuable, real-time snapshot of pricing dynamics in the Arabic market, offering immediate insights that studies may miss.

2. **Potential Biases** Our project accounts for several potential biases, including platform bias (as Shaguf is the only platform in focus), category bias (due to certain course categories being presented more), and measurement bias (errors in price extraction). By addressing these biases, we make sure that our findings are more reliable and reflect of current market conditions.

3. Data Sources

3.1 Data Source Description

Primary Data Source: Shaguf Platform, Shaguf is an online learning platform based in Saudi Arabia that offers a wide range of courses across various subjects, including technology, business, languages, and personal development. The platform primarily helps to Arabic-speaking learners and provides both free and paid courses. To encourage enrollment, it often offers discounts on course prices.

Why Shaguf?:

- **Local Relevance:** Shaguf offers valuable insights into the Arabic e-learning market, which is often not explored enough in global studies.
- **Rich Metadata:** The platform provides structured data such as price, original price, discount percentage, course duration, and difficulty level. It also includes unstructured data like instructor details and course descriptions.
- **Discount Transparency:** Shaguf clearly displays both the original and discounted prices, making it easy to calculate the exact discount percentage.
- **Public Accessibility:** All course pages are publicly accessible, which allows for ethical web scraping without the need for authentication or violating any terms of service.

Data Collection Method Data will be collected through web scraping using Python libraries like BeautifulSoup and Selenium. The process will involve extracting data from publicly available course listing pages and individual course detail pages.

Scraping will be done responsibly, with appropriate delays between requests to ensure that the platform's server is not overloaded, adhering to ethical scraping guidelines.

3.2 Details on the Data to be Collected

Feature	Data Type	Description	Example
title	String (Nominal)	Course title	"مقدمة في بايثون"
price	String → Float	Current price with currency symbol (to be converted)	"رس 199"
old_price	String → Float	Original price before discount (to be converted)	"رس 299"
discount_percent	Integer	Discount percentage applied	33
duration	String → Integer	Course duration in hours (to be converted)	"28 hours"
students	Integer	Number of enrolled students	1,234
rating	Float	Average user rating (0–5 scale)	4.5
level	String	Course difficulty level	"مبتدى"
instructor	String	Instructor name	"د. أحمد السالم"
university	String	Partner university (if applicable)	"جامعة الملك سعود"
has_discount	Boolean	Whether a discount is applied	True / False
final_price	Float (Computed)	Price after discount (if applicable)	199.0

Expected Sample Size:

- Approximately **~1,000 courses** across multiple categories and levels.
- Data will be collected in **February 2026** to ensure temporal consistency.

Note: Rating was extracted when available; however, due to limited variability across courses, it was not used as a primary analytical variable.

3.3. Bias & Limitations

3.3.1 Representation Bias

Question: Are all relevant groups fairly represented?

The dataset may suffer from representation bias in the following ways:

Bias Type	Description	Mitigation
Platform bias	Shaguf is a single platform; findings may not generalize to global platforms (Coursera, Udemy, edX).	Acknowledge this limitation and avoid overgeneralizing findings.

Category bias	Some categories (e.g., technology, business) may have more courses than others (e.g., arts, humanities).	Report category distribution and conduct category-wise analysis.
Level bias	Beginner-level courses may outnumber intermediate and advanced courses.	Compare pricing patterns across levels.
Language bias	Courses are primarily in Arabic; findings may not apply to non-Arabic markets.	Clearly define the scope as Arabic e-learning market.

3.3.2 Measurement Bias

Question: Are there biases in how data was measured or collected?

Bias Type	Description	Mitigation
Price extraction	Prices are extracted from text (e.g., "199 ر.س") and converted to numbers. Currency symbols or formatting inconsistencies may cause errors.	Use regex patterns carefully; manually validate a random sample.
Discount percentage	Some courses may not display old_price, making discount percentage calculation impossible.	Only include courses with complete price information in discount analysis.
Scraping timing	Prices and discounts may change over time. Scraping at a single point may capture temporary promotions.	Note the collection date in the dataset; avoid interpreting as long-term trends.

3.3.3 Historical Bias

Question: Does the data perpetuate historical injustices or inequalities?

Bias Type	Description	Mitigation
Pricing inequality	Courses from well-known universities or instructors may be priced higher regardless of quality, perpetuating prestige-based pricing.	Acknowledge this as a market reality; avoid value judgments.
Discount patterns	Discounts may be applied more frequently to certain categories (e.g., technology) while others receive fewer discounts, potentially reinforcing subject hierarchies.	Report discount patterns transparently without assuming causation.
Instructor reputation	Established instructors may command higher prices, potentially disadvantaging new or lesser-known educators.	Include instructor as a variable but avoid overinterpreting individual effects.

3.3.4 Additional Biases

Bias Type	Description	Mitigation
Survivorship bias	Only currently available courses are scraped; courses that have been removed are not included.	Acknowledge this limitation; the dataset represents the current market state.
Temporal bias	Data is collected at a single point in time, capturing only current prices and discounts.	Avoid claims about trends over time; frame as a snapshot analysis.

3.4 Summary of Dataset Characteristics

Aspect	Details
Source	Shaguf platform (web scraping)
Sample size	~1,000 courses
Collection date	February 2026
Data types	Structured (numeric, categorical) + Unstructured (course descriptions, instructor name, title)
Key variables	price, old_price, discount_percent, duration, students, level.
Primary use	Analyze relationships between duration, price, and discount strategies
Limitations	Single platform, Arabic-language focus, cross-sectional (snapshot) data

5. Objectives

The primary objective of this project is to analyze the factors influencing student enrollment across online courses on the Shaguf platform using scraped course level data. This study focuses on identifying measurable attributes that may be associated with higher or lower enrollment levels. Specifically, the objectives are:

- To examine the relationship between the number of enrolled students and course duration (in hours).
- To analyze how the final price of a course relates to student enrollment levels.

- To investigate whether the discount percentage is associated with higher student participation.
- To compare enrollment patterns across different course levels (Beginner, Intermediate, Advanced).
- To explore whether instructor or university reputation is linked to differences in enrollment.
- To identify which course attributes show the strongest association with student enrollment. By focusing on variables such as course duration, price, old price, discount percentage, course level, instructor, and the number of students, this project aims to provide data driven insights into pricing strategies and enrollment behavior within the Shaguf online learning environment.

6. Method

Data for this project was collected using a Python based web scraping script, executed in Google Colab. The scraping process was designed to extract publicly available course level data from multiple pages on the Shaguf platform. The following course attributes were collected:

- Number of students
- Course duration
- Current price
- Original price (old price)
- Discount percentage
- Course level (e.g., Beginner, Intermediate, Advanced)
- Instructor
- University
- Rating

Note that the rating values were also extracted, but due to limited variability across courses, they were not selected as a primary analytical variable. The extracted data was stored in CSV format to maintain its original structure.

Numeric-related fields, such as course duration and pricing, were preserved in their raw textual format before further processing to maintain data integrity. Pagination handling and controlled request timing were applied to ensure proper retrieval of course listings while maintaining ethical scraping practices and following web scraping guidelines.

7. Challenges & Recommendations

Challenges:

Several challenges were encountered during the data collection process:

- Some course attributes, like duration and pricing, were embedded in text format, which required careful extraction and conversion.
- Discount information was not uniformly structured across all course pages, making it difficult to extract this data consistently.
- The platform does not provide a public API, so data extraction had to be done manually via web scraping.
- Some courses had incomplete fields, such as missing old price values, which made certain analyses difficult.
- Pagination handling was necessary to retrieve courses across multiple listing pages, which added complexity to the data collection process.

Recommendations :

To improve future data collection and analysis, the following recommendations are made:

- Implement automated validation checks during scraping to detect and flag missing or inconsistent values.
- Maintain a timestamped dataset to track pricing and discount changes over time, allowing for a better analysis of trends.
- Expand data collection to include additional platforms to reduce platform specific bias and offer a more general point of view on pricing strategies and learner behavior.

Phase 2

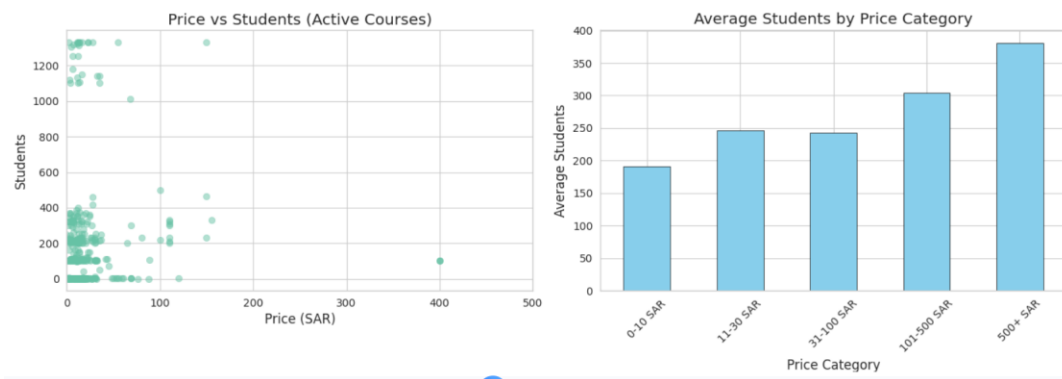
8. Primary Data Analysis (Shaguf)

To evaluate the current market standing and internal performance of Shaguf, we conducted a comprehensive Exploratory Data Analysis (EDA) on the platform's primary dataset. This analysis serves as the baseline for our comparison against global giants like Udemy and Coursera.

Challenges in Primary Collection: The primary dataset required significant cleaning to handle missing values in "Instructor Ratings" and "Duration." We also had to standardize the "Level" categories to ensure they were comparable with the international datasets used in our secondary analysis.

Context & Strategic Overview: Shaguf is a specialized academic platform that targets university students with a low-cost, high-accessibility courses. Unlike global platforms, Shaguf focuses on learners that seek academic content. An important finding is the importance of instructor reputation pattern, where students prioritize instructor credibility over price or discounts which reflects a trust-based approach rather than a what is believed that duration or discounts affect price.

Figure 8.1: Shaguf Price Distribution (Local Market)

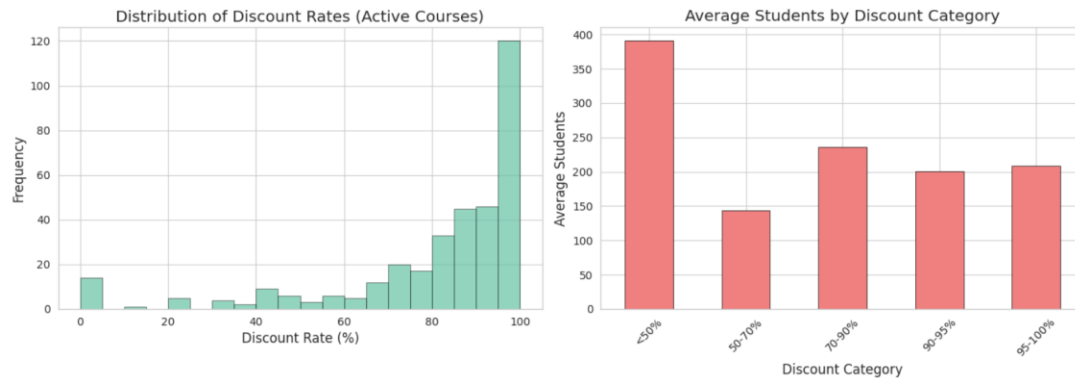


Description: Illustrates the spread of course prices across the Shaguf platform in SAR.

Observation: The chart shows that the price distribution is heavily skewed toward low-price courses, with 80% of active courses priced under 30 SAR

Insight: A small number of premium courses exist (3% above 500 SAR from the test sample), but their success is driven by outlier cases.

Figure 8.2: Discount Rate vs. Students (Active Courses)

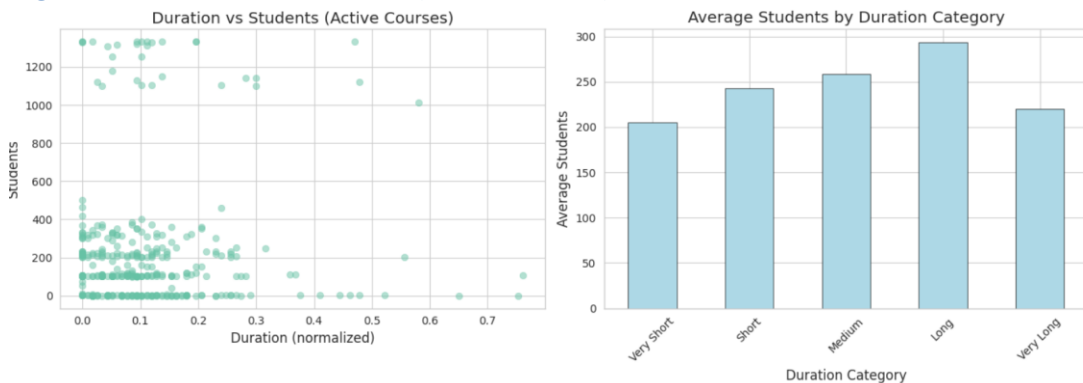


Description: This Bar plot illustrates the relationship between the discount percentage offered on courses and the resulting number of enrolled students.

Observation: Higher discounts (those exceeding 50% or 60%) do not show a direct correlation with an increase in student enrollment. In fact, courses with modest discounts or none at all often perform better.

Insight: Heavy discounting is not a primary driver for course success on the platform. Students appear to prioritize factors other than the magnitude of the price reduction when choosing a course.

Figure 8.3: Duration vs. Students (Active Courses)

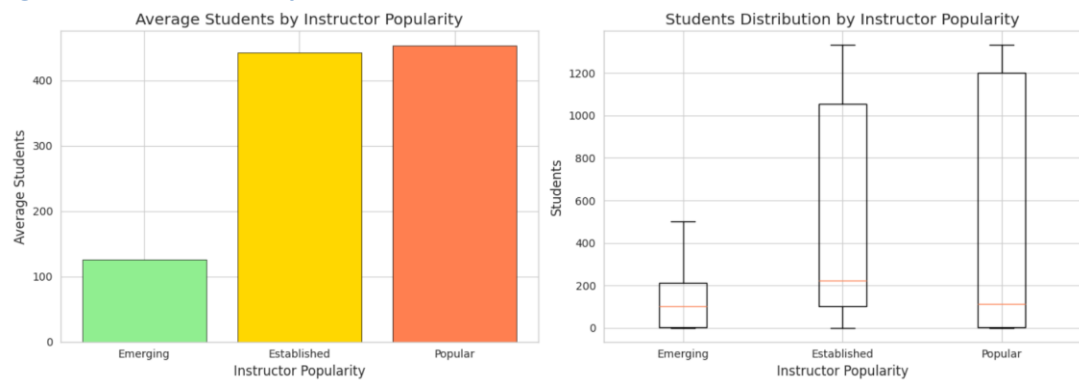


Description: This chart maps the duration (length) of active courses against the total number of students enrolled in them.

Observation: There is no significant correlation between the length of a course and its popularity. Both short and long courses show varied levels of enrollment without a clear trend line.

Insight: Course duration is a neutral factor. The Shaguf platform is dominated by short, focused content, and students do not seem to favor longer courses over shorter ones, or vice versa.

Figure 8.4: Instructor Reputation vs. Student Enrollment



Description: This bar chart compares the average or total student enrollment across three levels of instructor reputation: Emerging, Popular, and Established.

Observation: Established and Popular instructors have approximately 3.5 times more students than Emerging instructors.

Insight: Instructor reputation is the single most important driver of enrollment. This factor outweighs numerical variables like price or discount, as students tend to follow and trust experts with a proven track record

Limitations of Primary Data:

Incomplete Course Level Data: Course level information (Beginner, Intermediate, Advanced) was missing for 27.6% of courses. Which prevented a full analysis of how course difficulty is related to enrollment patterns.

Temporal Limits: The dataset provides a snapshot of current prices; it does not capture seasonal shifts or promotional cycles, making it harder to measure price change over time.

9. Secondary Data Analysis (Udemy & Coursera)

To support our primary analysis of the Shaguf platform, we needed secondary data from global platforms. We searched Kaggle and found two datasets that fit our needs:

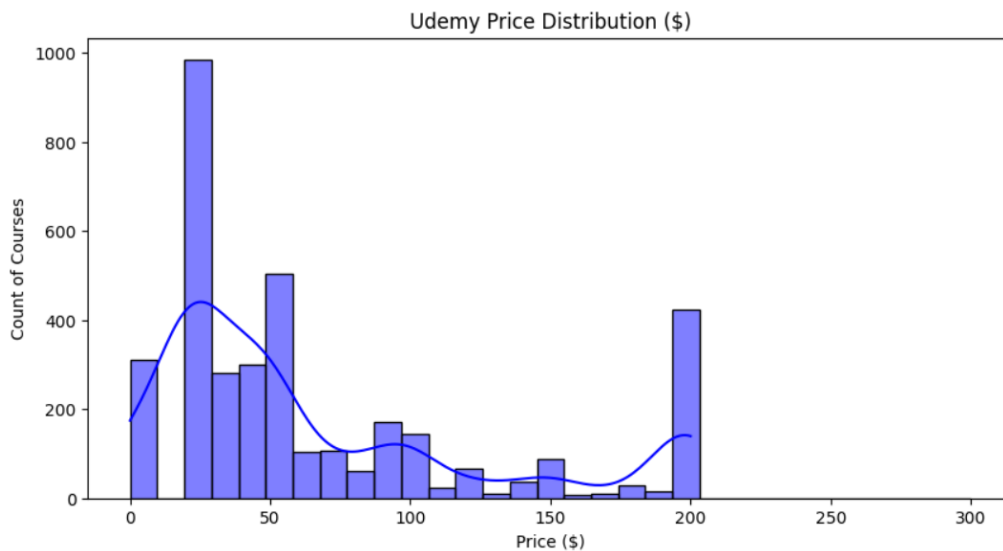
- **Udemy Courses Dataset**
- **Coursera Course Dataset**

The main challenge we faced was finding datasets that had similar features to our primary data, such as price, duration, and course level. After reviewing several options on Kaggle, we selected these two datasets because their structure was our closest match, allowing us to make a proper comparison.

To compare Shaguf with global platforms, we analyzed secondary datasets from Udemy and Coursera containing more than ~4,500 courses.

Our EDA reveals two special global strategies: Udemy's Marketplace Model, that depends on high price flexibility (\$20–\$200) and volume, and Coursera's Subscription Model, that usually would offer courses at \$0 for test. An important finding is the "Duration Anomaly"; our correlation analysis shows that global pricing follows an approach that prefers value, as course length showed a close to zero correlation with price.

Figure 9.1: Global Price Distribution (Udemy)

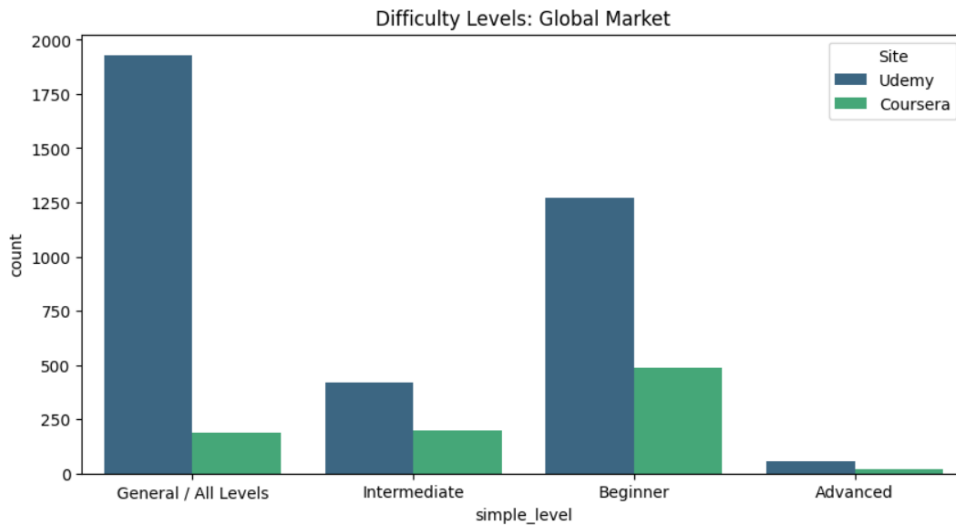


Description: Illustrates the concentration of course prices on Udemy.

Observation: The distribution is Right-Skewed, with a significant concentration in lower price brackets \$20 - \$200.

Insight: learners globally are highly price sensitive. Udemy's frequent discounting is an important entry barrier for any new competitors.

Figure 9.2: Global Difficulty Levels (Udemy vs. Coursera)

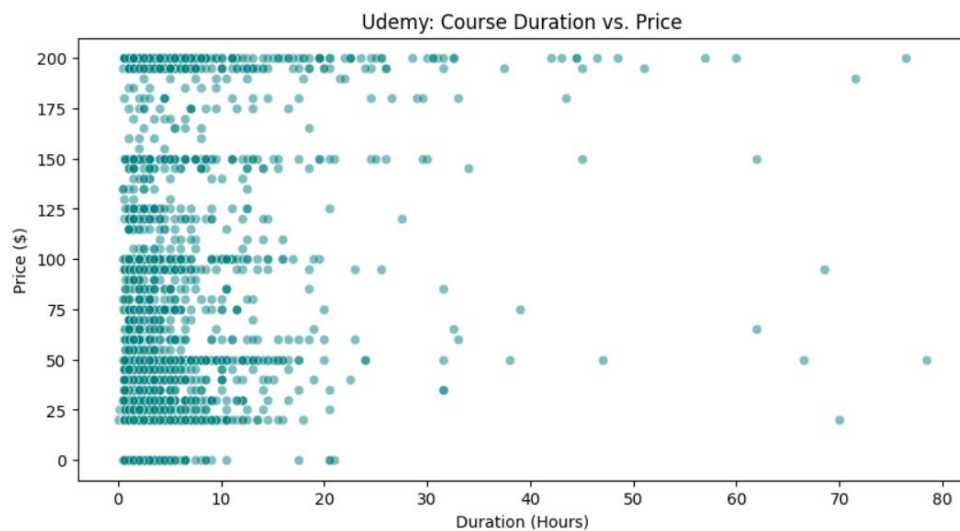


Description: Compares target audience levels across global platforms.

Observation: Beginner-level courses dominate the market > 60%.

Insight: Global platforms prioritize General Appeal. This "General" trend should leave a gap for specialized academic platforms like Shaguf.

Figure 9.3: Udemy: course duration vs price



Description: Shows the correlation of a course duration and its price.

Observation: clearly shows a near no correlation between how long a course is and its price.

Insight: The correlation is weak. This confirms our study's idea that price is not just about how long the course is, but about other strategic factors.

Limitations of Secondary Data

Data Gaps: The Coursera dataset lacked obvious "Pricing" information, limiting the duration and price correlation analysis primarily to Udemy.

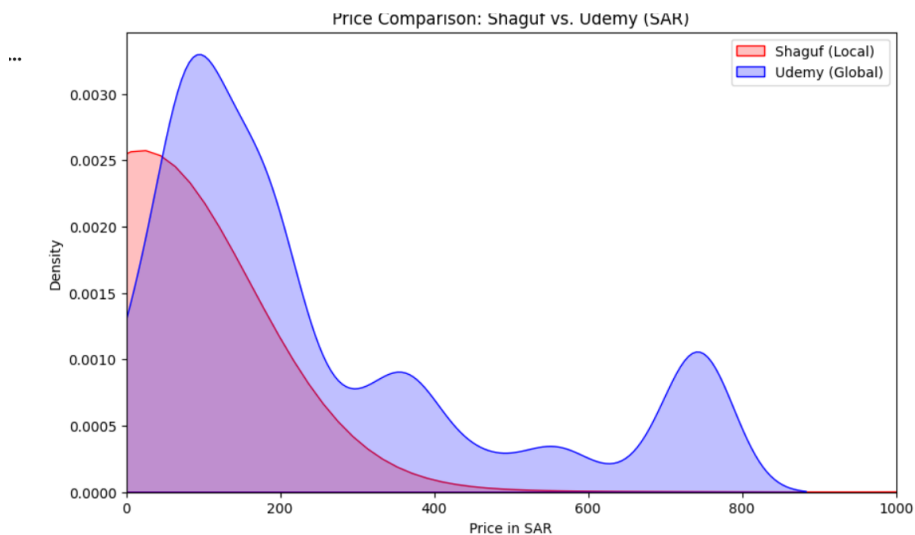
Currency Conversion: Prices in the dataset were originally in USD, so they were converted to SAR for comparison. but, this conversion may not fully reflect differences.

10. Comparative Analysis: Local vs. Global Patterns

In this section, we highlight the similarity and differences between **Shaguf** (Local) and global platforms **Udemy & Coursera**. This comparison should validate our primary findings against secondary data findings.

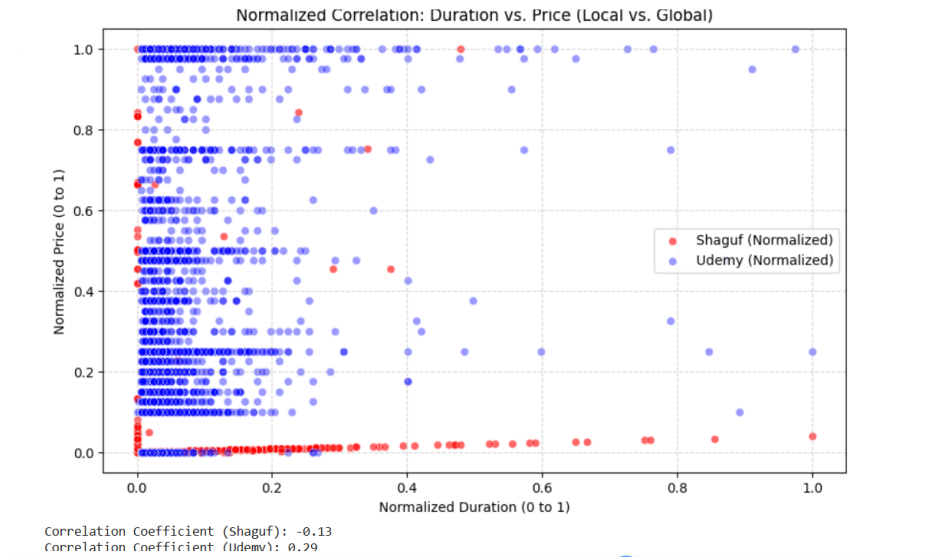
10.1 Pricing similarity & Distribution

After converting prices to SAR, we found that Shaguf's pricing is more "clustered" (focused on academic), while Udemy shows a broader, wider distribution. Both platforms utilize discounts as a primary reason for enrollment.



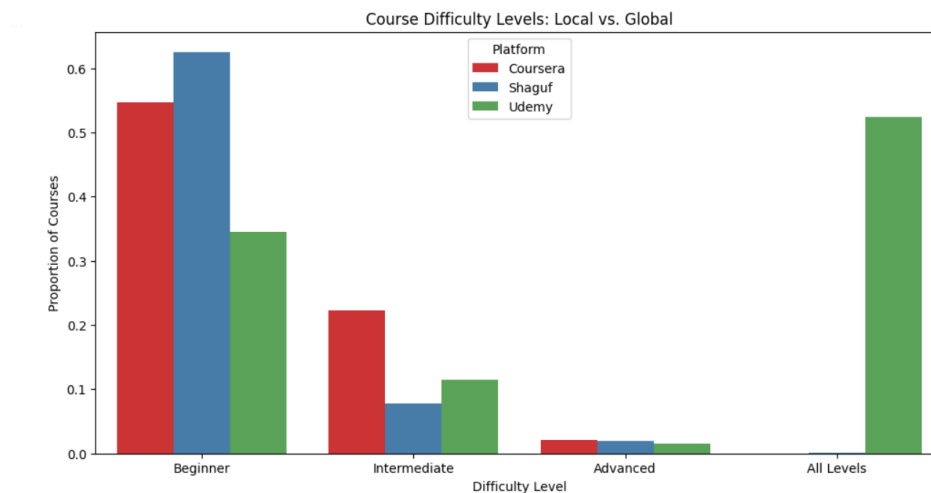
10.2 Normalized Pattern Comparison (Duration vs. Price)

By applying Normalization (Scale 0 to 1) to both Shaguf and Udemy, we compared the price and duration patterns. We found that both markets follow the same logic: course duration has no notable impact on the final price. This confirms that Shaguf's pricing strategy is on theme with global industry, where value is driven by how important the topic is rather than its length.



10.3 Difficulty Levels Comparison

Using Shaguf's primary level data, we found a significant difference. While global platforms like Udemy and Coursera focus more on Beginner courses over 60% of its content, Shaguf provides a sparse selection of Intermediate and Advanced academic courses. This shows that Shaguf's value is focused on academic content rather than general skill discovery (like Beginner courses).



11. Summary of Insights & Future Hypotheses

11.1 Key Insights

- **Course Duration Is Not the Main Factor for Enrollment**

The analysis suggests that the length of a course does not always determine how many students enroll in it. Some shorter courses still attract a good number of students. This shows that students may care more about how useful and relevant the course is rather than the total number of hours.

– **Shaguf Focuses on Academic-Level Courses**

The distribution of courses on the platform shows that many of them are related to university subjects and are offered at intermediate or advanced levels. This suggests that Shaguf is positioned as a platform that supports academic learning rather than general skill exploration.

– **Instructor Presence May Influence Enrollment**

The analysis also indicates that courses taught by instructors who have a stronger presence on the platform tend to have higher enrollment numbers. This suggests that instructor reputation or familiarity may influence students when choosing a course.

11.2 Proposed Hypotheses

As we move forward, we suggest testing the following hypotheses through further statistical analysis:

Hypothesis 1 – Course Duration and Enrollment

Increasing the total number of hours in a course on Shaguf does not necessarily lead to higher student enrollment.

Rationale

The exploratory analysis suggested that some shorter courses were able to attract a considerable number of students. This indicates that students may prefer courses that are more focused and directly related to their academic needs rather than courses that simply contain more hours of content.

Hypothesis 2 – Course Level Influence

The level of a course may influence either the course price or the number of students enrolling in the course.

Rationale

Since Shaguf mainly provides courses related to academic subjects, the course level may affect how students evaluate the course value. Intermediate or advanced courses may be perceived as more specialized and therefore may influence enrollment behavior or pricing decisions.

Hypothesis 3 – Instructor Reputation Effect

Courses taught by well known or established instructors may attract higher enrollment compared to courses taught by emerging instructors.

Rationale

The exploratory analysis indicated that instructors with a stronger presence on the platform tend to have courses with higher student enrollment. This suggests that instructor reputation may play an important role in student trust and course selection.

Model 1: Baseline Model – Linear Regression

Model Objective

Linear Regression was used as the first regression model to predict student enrollment. This model was selected as the baseline because it is simple, easy to interpret, and gives us a starting point for comparing the performance of more advanced models.

Dataset and Features

The model was trained using the combined Shaguf and Udemy dataset. The target variable was `students_log`, and the selected features were `price_log`, `is_free`, and `has_discount`.

These features were chosen because they are simple and directly related to the project goal of studying how course pricing and discounts relate to student enrollment.

Model Training

The data was split into 80% training and 20% testing. We used Linear Regression without extra tuning because the purpose of this model was to create a simple benchmark, not to build the strongest model.

Performance

The Linear Regression baseline model achieved:

Metric	Result
MAE	1.6617
RMSE	2.0488
R²	0.1613

Since `students_log` was used as the target variable, MAE and RMSE are interpreted on the log-transformed scale, not as the exact number of students.

Visual Evaluation

The Actual vs Predicted plot showed that the predictions were scattered and did not closely follow the actual values. The Residual Plot also showed that the errors were not fully random around zero.

This means that the simple Linear Regression model could not fully capture the relationship between the selected course features and student enrollment.

Model Interpretation

The baseline model showed limited performance, with an R^2 score of 0.1613. This means that the model explained only a small part of the variation in student enrollment.

This result was expected because the baseline model used only a few simple features. However, it still provides an important reference point for comparing the Decision Tree and Random Forest models later.

Model 2: Random Forest Regressor

Model Objective

Random Forest Regressor was used as the second regression model to predict student enrollment.

This model was used because it can handle more complex relationships compared to the baseline model.

Dataset and Features

The model was trained using the combined Shaguf and Udemy dataset.

The target variable was `students_log`, and the selected features were `price_log`, `is_free`, `has_discount`, and `duration`.

Model Training

The data was split into 80% training and 20% testing.

We used `n_estimators = 200` and `max_depth = 6` to keep the model balanced and reduce overfitting.

Performance

The Random Forest model achieved:

Metric	Result
MAE	1.7913
RMSE	2.2179
R²	0.4827

Since `students_log` was used as the target variable, MAE and RMSE are interpreted on the log-transformed scale, not as the exact number of students.

Feature Importance

The most important feature was `has_discount`, followed by `price_log` and `duration`.

This means that discount had the strongest impact, while price and duration also played a role in the predictions.

Overall, the model performed better than the baseline and was able to explain a larger part of the variation in student enrollment

Model 3: Decision Tree Regressor

Model Objective

Decision Tree Regressor was used as the third regression model to predict student enrollment. This model was used because it can capture non-linear relationships between course features and enrollment.

Dataset and Features

The model was trained using the combined Shaguf and Udemy dataset. The target variable was `students_log`, and the selected features were `price_log`, `discount_rate_actual`, `is_free`, `has_discount`, and `source_Udemy`.

Model Training

The data was split into 80% training and 20% testing. We used `max_depth = 5` to keep the tree simple and reduce overfitting.

Performance

The Decision Tree model achieved:

Metric	Result
MAE	1.6235
RMSE	2.0223
R²	0.1828

Since `students_log` was used as the target variable, MAE and RMSE are interpreted on the log-transformed scale, not as the exact number of students.

Feature Importance

The most important feature was `price_log`, followed by `discount_rate_actual`. This means that price and discount rate were the main features used by the Decision Tree model when predicting student enrollment.

Model Comparison and Best Model Selection

After building the three regression models, we compared their performance using MAE, RMSE, and R^2 . All models were trained and evaluated using the same combined Shaguf and Udemy dataset, so the comparison would be fair.

Model	MAE	RMSE	R^2
Linear Regression	1.6617	2.0488	0.1613
Decision Tree	1.6235	2.0223	0.1828
Random Forest	1.7900	2.2100	0.4800

The **Linear Regression model** was used as the **baseline**. Its R^2 score was **0.1613**, which means it explained only a small part of the variation in student enrollment.

The **Decision Tree model** performed slightly better than the baseline. It had the lowest MAE and RMSE, which means its average prediction error was the smallest among the three models. However, its R^2 score was still low at **0.1828**.

Random Forest had higher MAE and RMSE than the other two models, but it achieved the highest R^2 score, which was **0.48**. This means it explained the largest amount of variation in student enrollment compared to the other models.

Based on this comparison, **Random Forest was selected as the best overall model**. We selected it because it had the strongest R^2 score, meaning it captured the general pattern in the data better than the other models. Even though Decision Tree had lower prediction error, **Random Forest gave a better overall explanation of student enrollment patterns.**

Overall, the results show that tree-based models performed better than the simple Linear Regression baseline. The Random Forest model was the most suitable final model for our prediction task.

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